[1] Curriculum Vita

Name: **Adabala Venkata Snehith**

Interests: Computational Chemistry & Materials Science

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Website: `https://snehithavchem.github.io/AND`

Currently as: Integrated Ph.D (2023–) at Tata Institute of Fundamental Research, Hyderabad



Education:

Certificate/Degree	Year	University/Board	CGPA/Percentage
10th	2017	Telangana State Board of School Education	10.0/10.0
12th	2019	Telangana State Board of Intermediate Education	97.1%
B.Sc	2023	Pondicherry University	9.42/10.0

Research Experience:

1. National Initiative for Undergraduate Sciences (NIUS) - Project Student

Conducted by: Homi Bhabha Centre for Science Education (HBCSE), Tata Institute of Fundamental Research (TIFR), Mumbai.

Completed: May, 2022

Project Title: A Detailed Study on Metal-Chalcogenide Based Photocatalysts for CO_2 Photoreduction

Project Mentor: Dr. Dimple P Dutta, SO(G), BARC Mumbai

2. International Centre for Clean Water (ICCW) - Research Intern

Conducted by: Indian Institute of Technology, Madras

Completed: March, 2023

Project Title: A Detailed Study on Machine Learning approach for effective detection of heavy metal ions via Electrochemical Sensors

Project Mentor: Dr. Ganapati Natarajan, Principal Scientist, Modelling & Simulation Vertical

[1] As of 10th Oct, 2026

3. Visiting Summer Research Programme - Visiting Student

Conducted by: Tata Institute of Fundamental Research, Hyderabad

Completed: August, 2023

Project Title: Computational Studies on the Mechanism of Epoxide Ring Opening of Styrene Oxide on Acidic Alumina Silica (AAS)

Project Mentor: Dr. Soumya Ghosh, TIFR Hyderabad

4. Lab Rotation Project : Project-A student

Conducted by: Tata Institute of Fundamental Research, Hyderabad

Completed: May, 2024

Project Title - 1: Understanding spectra of molecules confined in optical cavities

Project Mentor -1 : Dr. Pranav R Shirhatti, TIFR Hyderabad

Project Title - 2: Effects of Surface Termination in Type-II Heterojunctions

Project Mentor - 2 : Dr. Soumya Ghosh, TIFR Hyderabad

Conferences/Courses Attended:

- *Attended Conference:* Presented Poster Conference on Advancements in Renewable Energies (CARE-2026), Mumbai from September 23 - 25, 2026.
- *Attended Conference:* Presented Poster at TIFR Biennial Chemistry Conference, Hyderabad from August 06 - 08, 2026.
- *Attended Conference:* Presented Poster at Theoretical Chemistry Symposium - 2025 at Indian Institute of Technology, Mumbai from December 02 - 05, 2025.
- *Attended School:* International School and Conference on Evolution of Electronic Structure Theory Experimental Realization held at SRM Chennai & Indian Institute of Technology, Madras from January 03 - 07, 2025.
- *Attended Course:* International Winter School: Frontiers in Materials Science held at Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bengaluru, India from December 06 - 10, 2021.
- *Attended Course:* Summer Course on "Curves and Surfaces: Geometry and Physical Applications" in online mode held at International Centre for Theoretical Sciences (ICTS), Tata Institute of Fundamental Research(TIFR), Bengaluru, India, from May 17 to June 15, 2022, by Prof. Joseph Samuel, ICTS-Simon Visiting Professor, ICTS Bengaluru.

Papers Published:

Research Articles:

3. P.De., *S, Adabala* et al., **Unprecedented HAT rate acceleration in water by a non-heme manganese oxo complex**, *Chem. Sci.* 2026, 17, 4042-4049.

2. S,Anil., D,Maity., *S, Adabala* et al., **Valley Engineering of Monolayer MoS_2 via Substitutional Cr and V Dopants and the Mechanistic Insights**, *Nano Lett.* 2025, 25, 35, 13401–13409.

1. R, Hammad., *S, Adabala.*, S, Ghosh., **Choice of Layering and Band Alignment in 2D Heterostructures**, *J. Phys. Chem. C* 2024, 128, 48, 20635–20640.

Review Articles:

2. S.Gopinath., *S, Adabala.*, et al., **Structural, Electronic, and Interfacial Pathways Governing the Stability of $LiMn_2O_4$ Spinel Cathodes**, *Batt. & Supercaps.* 2026, 9, 2, e202500742.

1. *S, Adabala.*, D.P, Dutta., **A review of recent advances in metal chalcogenide-based photocatalysts in CO_2 Photoreduction**, *J.Env.Chem.Eng.* 10,107763 (2022).

Computer Skills:

Linux (Ubuntu) & Windows, BASH Scripting, Python, Fortran, MATLAB, Gaussian, VASP

Other activities:

- Student Member of American Association of Cancer Research (AACR) from April 2023 (ID: 127032).
- Associate Member at the International Society of Development and Sustainability (ISDS) from March 2022.
- Appreciation Honour from Royal Academy of Music, Great Britan for notable performance in *Bansuri* in 2022.
- A Course Diploma in Fashion Designing from National Institute of Fashion Technology (NIFT, Mumbai) in 2023.

Personal Details:

Gender,Nationality, Marital Status: Male, Indian, Single

Date of Birth: 18.07.2002

Languages: English, Hindi, Telugu, Tamil, Malayalam, Marathi